B E N C H M A R K R E P O R T

When the Answer Is in the Question And Two Models Still Miss It

What Easy Questions Reveal About Small Model Limitations

Behavioral Analysis of 22 LLMs on a Primary Clarifier Design Question

22	**86.4%**	**19/22**	**10th**
LLMs Tested	Reported Pass Rate	Chose Covered Top	R1 Parsing Issue

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1. Executive Summary

I posed a primary clarifier design question to 22 LLMs: a wastewater treatment plant is located 300 meters from a residential area. The plant has an odor collection and treatment system. What is the most important design criterion for the primary clarifier? The correct answer is a covered top, which enables odorous gases to be captured and routed to the treatment system. The trap consists of shape options (circular, rectangular, square) that appeal to conventional design thinking but do not address the odor constraint.

The reported pass rate is 86.4% (19/22). The adjusted rate, accounting for DeepSeek R1's parsing issue, is 90.9% (20/22). This places the benchmark in the easy tier alongside Material Selection (100%) and Sample Preservation (95.5%). After twelve benchmarks, the cumulative funnel remains at zero: 22 → 17 → 15 → 2 → 0 → 0 → 0 → 0 → 0 → 0 → 0 → 0.

F1 — Easy Question, Revealing Failures: Only two models genuinely failed: Gemma 1B and Gemma 4B. Both are small Gemma models. Every other model — including all Claude, Gemini, Qwen, and DeepSeek variants — chose correctly. When a question is easy enough that 20 of 22 models pass, the two failures become more informative than the twenty successes.

F2 — Gemma 1B Chose the Dangerous Opposite: Gemma 1B selected F (open top) — the exact opposite of the correct answer. Its reasoning stated that "open tops can lead to increased odor emissions" and then chose the open-top option anyway. The answer is internally contradictory: the model acknowledged the problem and selected the option that maximizes it. This is the third benchmark where Gemma 1B has produced a dangerous or contradictory answer.

F3 — Gemma 4B Fell for the Shape Trap: Gemma 4B selected B (circular shape) and explicitly dismissed the covered-top option: "Helps contain odors within the tank, but doesn't fundamentally reduce their generation." The model reframed the question from odor containment to odor reduction — a subtle but critical misunderstanding. The question asks about capturing existing odors for treatment, not preventing their formation.

F4 — R1: Tenth Parsing Issue in Twelve Benchmarks: DeepSeek R1 answered E (correct) in a 3,710-character think block. The parser read "T" from the <think> tag. This is R1's 10th parsing issue in 12 benchmarks (83%). On this question, R1's reasoning was straightforward and correct — the loss is entirely formatting.

F5 — Gemini 2.5 Flash: Third One-Token Answer: Gemini 2.5 Flash responded with a single character: "E." This is its third one-token response in the series. Across six benchmarks, its pattern is emerging: when the model produces a short answer, it tends to be correct; when it produces a long answer, it can be wrong (468 tokens on Odor Pipe #11 → FRP trap).

2. The Question

The scenario presents a common WWTP design situation: a primary clarifier near a residential area with an existing odor collection system. The question tests whether models can identify that covering the clarifier is essential for integrating it with the odor collection infrastructure — rather than focusing on the geometric shape of the tank.

The trap is structural: four of six options describe tank shapes (A–C) or configuration (D), while only two options address the top design (E = covered, F = open). The prompt explicitly states that the plant has an odor collection system, creating a logical chain: odor complaints → cover the tank → connect to collection system → route to treatment. Models that follow this chain arrive at E; models that anchor on conventional clarifier design criteria (shape, hydraulics) choose differently.

With 6 options, the random correct answer probability is 16.7%.

Table 1: The Six Options

Option	Description	Type	Rationale
A	Square shape	Distractor	Uncommon in practice. No impact on odor control.
B	Circular shape	TRAP	Hydraulically efficient — a legitimate design preference but irrelevant to the odor constraint.
C	Rectangular shape	Distractor	Easier to construct, common for large plants. No impact on odor control.
D	Connect clarifiers	Distractor	Operational flexibility. Unrelated to odor management.
E	Covered top	CORRECT	Captures odorous gases. Enables connection to odor collection system for treatment.
F	Open top	DANGEROUS	The direct opposite of the correct answer. Guarantees odor complaints.

Table 1: The question includes both a trap (shape options that appeal to conventional design thinking) and an anti-answer (F, open top). The presence of F makes this question a test of basic reasoning coherence as well as domain knowledge.

3. Results

Figure 1: Answer Distribution

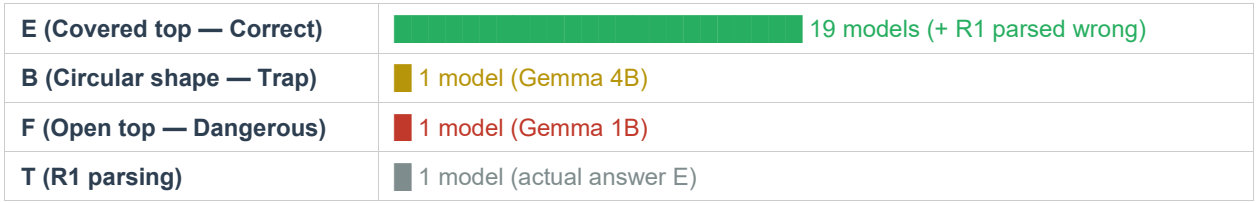

Figure 1: No model chose A, C, or D. The three failures split across three different wrong answers: shape (B), open top (F), and parsing (T). There was no clustering on any single wrong answer.

Table 2: Complete Results — All 22 Models

Model	Ans	Status	Out Tok	Latency	Family
Claude Haiku 4.5	E	PASS	292	4.0s	Claude
Claude Opus 4.1	E	PASS	245	10.6s	Claude
Claude Opus 4.5	E	PASS	316	10.2s	Claude
Claude Sonnet 4	E	PASS	165	5.4s	Claude
Claude Sonnet 4.5	E	PASS	295	8.7s	Claude
Gemini 2.0 Flash	E	PASS	239	3.3s	Gemini
Gemini 2.0 Flash Lite	E	PASS	106	2.3s	Gemini
Gemini 2.5 Flash	E	PASS	1	4.7s	Gemini
Gemini 2.5 Pro	E	PASS	175	14.1s	Gemini
Gemini 3 Flash Preview	E	PASS	132	5.6s	Gemini
Gemini 3 Pro Preview	E	PASS	185	12.5s	Gemini
Gemma 3 12B	E	PASS	~370	8.1s	Gemma
Gemma 3 4B	B	FAIL	~350	8.3s	Gemma
Gemma 3 27B	E	PASS	~350	8.0s	Gemma
Gemma 3 1B	F	FAIL	119	3.5s	Gemma
DeepSeek R1 (CoT)	E†	FAIL	885	6.2s	DeepSeek
DeepSeek V3.1	E	PASS	138	1.9s	DeepSeek
DeepSeek V3.2	E	PASS	40	2.6s	DeepSeek
Qwen3 235B A22B Instruct	E	PASS	155	2.1s	Qwen
Qwen3 Coder 480B	E	PASS	143	2.4s	Qwen
Qwen3 Next 80B Instruct	E	PASS	147	2.2s	Qwen
Qwen3 Next 80B Thinking	E	PASS	468	4.6s	Qwen

Table 2: †R1 answered E inside a <think> block. Parser read "T." This is R1's 10th parsing issue in 12 benchmarks (83%).

Parsing note: the Kaggle Benchmarks platform evaluates answers by extracting the first alphabetic character from each model's raw response. R1's <think> block causes the parser to read "T" instead of the intended answer. The reported pass rate (86.4%) reflects this rule; the adjusted rate (90.9%) reflects the actual answer content.

4. LLM Behavior Analysis

4.1 Two Failures, Two Different Mechanisms

The two genuine failures on this benchmark — Gemma 1B and Gemma 4B — failed for entirely different reasons. One chose the opposite of the correct answer; the other chose a plausible but irrelevant alternative. Together, they illustrate two distinct small-model failure modes.

Gemma 1B selected F (open top). Its reasoning contained the sentence: “Open tops can lead to increased odor emissions and potential for splashing.” This is a correct observation — open tops do increase odor emissions. The model then selected the open-top option as its answer. The response is internally contradictory: the argument against F is present in the reasoning that concludes with F. Additionally, the model recommended a rectangular shape within its justification for selecting F, suggesting confusion between the question being asked and the answer being given.

This is the third benchmark where Gemma 1B has produced a concerning answer. In Confined Space (#6), it called a dangerous response “the most logical and safest course of action.” In Odor Pipe (#11), it chose HDPE despite the UV requirement. The pattern suggests that Gemma 1B’s reasoning capability is insufficient for domain-specific questions where multiple constraints must be weighed simultaneously.

Gemma 4B selected B (circular shape). Unlike Gemma 1B, its reasoning was internally coherent — it argued that circular shapes reduce turbulence and therefore reduce odor generation. The model explicitly acknowledged and dismissed the covered-top option: “Helps contain odors within the tank, but doesn’t fundamentally reduce their generation.” This reveals a specific misunderstanding: the model framed the question as “how to reduce odor production” rather than “how to capture and treat existing odors.” The prompt explicitly mentions an odor collection system, making containment — not reduction — the relevant criterion.

4.2 Reasoning Quality Among Passing Models

Among the 19 models that chose E, reasoning quality varied significantly. Eighteen of 19 explicitly connected the covered top to the odor collection system — recognizing that covering is not just about containment but about enabling the existing infrastructure to function. Twelve of 19 explicitly noted that tank shape is a secondary consideration in this context. The remaining seven did not engage with the shape options at all — they went directly to the odor-cover-collection chain without discussing geometry, suggesting they never considered the shape distractor as relevant.

Token output ranged from 1 (Gemini 2.5 Flash) to 468 (Qwen3 Next 80B Thinking) among passing models, excluding R1. DeepSeek V3.2 provided the shortest reasoned answer at 40 tokens, capturing the essential logic in two sentences. Claude Sonnet 4 was the most concise among Claude models at 165 tokens. The range suggests that this question’s answer is well-represented in training data — models do not need extensive reasoning to arrive at it.

Gemini 2.5 Flash’s single-token response marks its third one-token answer in the series. Across six benchmarks analyzed for this model, a pattern appears to be emerging: its three shortest responses (1 token each) were all correct, while its longest response (468 tokens on Odor Pipe #11) was wrong. Whether brevity correlates with confidence and accuracy, or whether this reflects different internal processing paths, cannot be determined from this data alone — but the pattern is consistent enough to note.

4.3 R1 Parsing — Tenth Occurrence

DeepSeek R1 produced a 3,710-character think block that methodically evaluated each option, correctly identified odor control as paramount, and concluded with “E is the clear choice.” The parser read “T” from the <think> tag. This is R1’s 10th parsing issue in 12 benchmarks, raising the occurrence rate to 83%.

On the previous benchmark (#11), R1’s actual answer was wrong for the first time. Here, R1 returns to its established pattern: correct reasoning, lost to formatting. Across 12 benchmarks, R1 has now produced the correct answer in 8 cases but received credit for only 1 (Material Selection #1, where the parser happened to read the right character). The gap between R1’s reasoning quality and its scored performance is the largest of any model in the series.

Table 3: R1 Parsing History — Updated

Benchmark	Raw	Answer	Correct?	Issue
#1 Material Selection	<	B	Yes (PASS)	<think> block
#2 Pump RCA	<	A	Yes (FAIL*)	<think> → parsed wrong
#3 Dewatering RCA	<	A	No (FAIL)	<think> + wrong answer
#4 Biogas	—	—	—	(data pending)
#5 Walkway Material	*	B	No (FAIL)	**B** markdown
#6 Confined Space	*	O	Yes (FAIL*)	**O** → parsed wrong
#7 Sample Preservation	<	D	Yes (FAIL*)	<think> → T parsed
#8 Tool Selection	<	D	Yes (FAIL*)	<think> → T parsed
#9 Emergency Flocc.	<	C	Yes (FAIL*)	<think> → T parsed
#10 SCADA Cabling	<	E	Yes (FAIL*)	<think> → T parsed
#11 Odor Pipe	*	C	No (FAIL)	**Answer: C** → A parsed
#12 Primary Clarifier	<	E	Yes (FAIL*)	<think> → T parsed

Table 3: R1 has encountered formatting issues in 10 of 12 benchmarks (83%). In 7 cases, the correct answer was lost to formatting. #11 remains the only benchmark where R1 was genuinely wrong.

4.4 Family Performance — Recovery After the Storm

Figure 2: Family Pass Rates

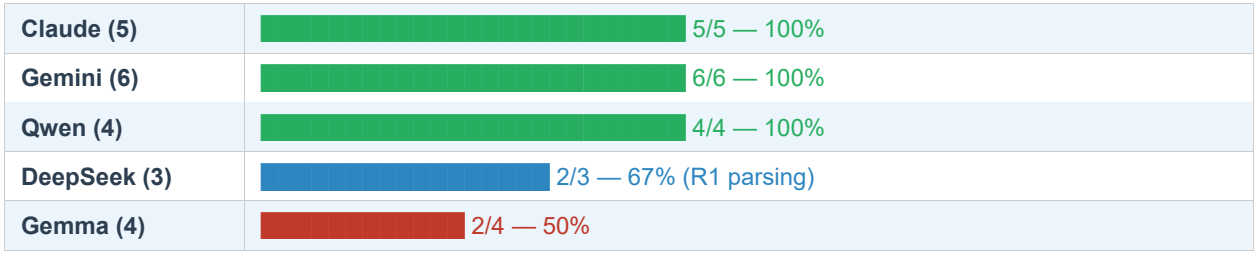

Figure 2: Three families at 100%. The only failures are Gemma’s two smallest models and R1’s persistent formatting issue.

After two consecutive benchmarks of significant family failures — SCADA Cabling (#10) broke Claude's streak, Odor Pipe (#11) saw no family above 40% — this benchmark represents a recovery. Claude returns to 100% after two benchmarks at 40%. Gemini returns to 100% after 33% on Odor Pipe. Qwen returns to 100% after its first-ever 0% on Odor Pipe. The same families that struggled on a hard question perform flawlessly on an easy one.

This contrast reinforces the series' central observation: family performance is question-dependent, not inherent. Claude is not "always reliable" or "recently unreliable" — it is reliable on questions within its knowledge boundary and unreliable on questions outside it. The same applies to every family.

Gemma remains the most variable family in the series. Its two small models (1B and 4B) fail here while the larger models (12B and 27B) pass — a straightforward scaling pattern on this benchmark. But this pattern has not been consistent across the series: Gemma 4B passed Tool Selection (#8) while Gemma 27B failed, and Gemma 4B passed Odor Pipe (#11) while Gemma 12B failed. Gemma's scaling behavior appears to be question-dependent rather than monotonic.

4.5 Trap Taxonomy Update — Twelve Benchmarks

Table 4: Updated Trap Taxonomy

Benchmark	Trap Name	Pass %	Failure Mode
#1 Material Selection	—	100%	
#2 Pump RCA	Electrical Trap	77%	Electrical anchoring
#3 Dewatering RCA	Local Trap	68%	Local anchoring
#4 Biogas	Aerobic Trap	14%	Aerobic anchoring
#5 Walkway	Corrosion Trap (FRP — Structural)	36%	Chemical resist. anchoring
#6 Confined Space	"Almost Safe" Trap	68%	Completeness illusion
#7 Sample Preserv.	Acid Trap	95.5%	Protocol anchoring
#8 Tool Selection	"False Safe" Trap	68.2%	Hazard blindness
#9 Emergency Flocc.	Coagulant Trap	68.2%	Mechanism confusion
#10 SCADA Cabling	Cost-Reliability Trap	63.6%	Partial knowledge
#11 Odor Pipe	UV Anchoring (FRP — UV)	27.3%	Judgment error
#12 Primary Clarifier	Shape Trap	86.4%	Shape anchoring / contradiction

Table 4: Twelve benchmarks. The Shape Trap is the least effective in the series — only two models fell for it, both from the smallest size category. This suggests the shape-vs-cover distinction is well-represented in training data.

5. Cross-Benchmark Comparison — Twelve Questions Deep

Figure 3: Pass Rate Across 12 WWTP Benchmarks

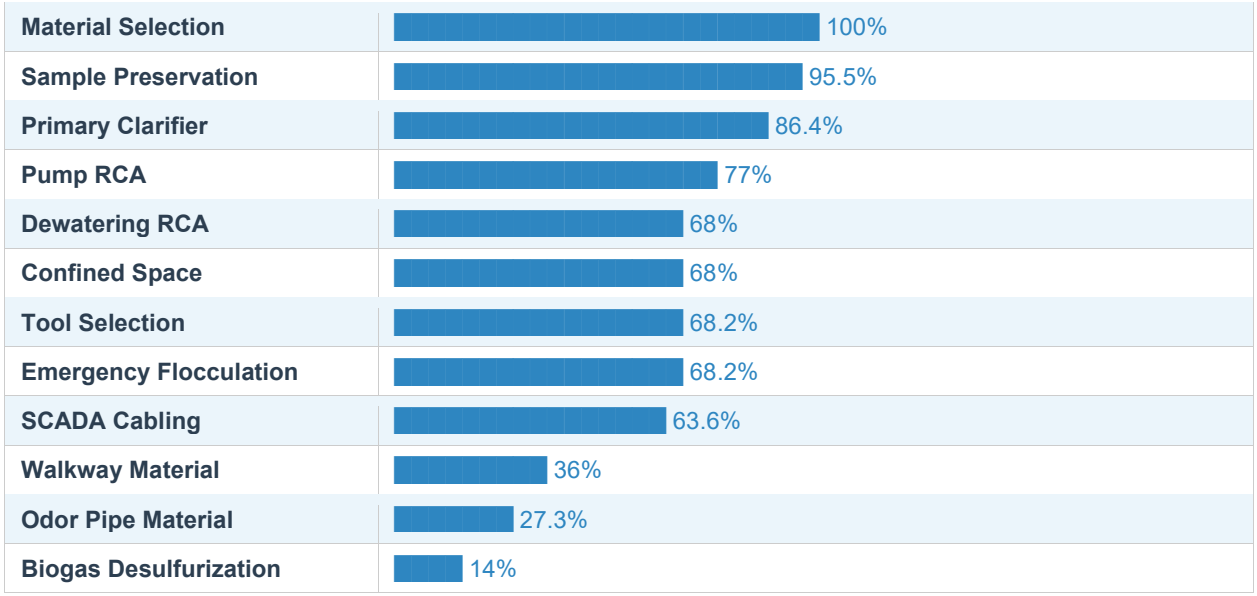

Figure 3: Twelve benchmarks spanning 14% to 100%. Primary Clarifier enters the easy tier at 86.4%, the third-easiest in the series.

The difficulty tiers are now clearly established with twelve data points. The easy tier (above 85%) contains three benchmarks where the answer is well-represented in training data and the logical chain is direct. The mid-range cluster (63–77%) contains six benchmarks that require domain-specific knowledge beyond common training data. The hard tier (below 40%) contains three benchmarks that require multi-criteria evaluation where the most salient criterion is not the decisive one.

The cumulative funnel remains at zero for the eighth consecutive benchmark: 22 → 17 → 15 → 2 → 0 → 0 → 0 → 0 → 0 → 0 → 0 → 0. The funnel closed at benchmark #5 (Walkway Material), which was the first hard-tier question in the series — suggesting that a single difficult question is sufficient to eliminate all perfect records. It has remained at zero through eight additional benchmarks of varying difficulty. No model has passed all twelve questions.

6. Conclusions

This benchmark adds a twelfth dimension to the WWTP series and provides a counterpoint to the previous two benchmarks' dramatic failures. At 86.4%, it demonstrates that when the logical chain is direct and the answer is well-represented in training data, the vast majority of models perform correctly. The failures that do occur are confined to the smallest models in the series.

Gemma 1B's selection of F (open top) — the exact opposite of the correct answer, contradicted by its own reasoning — is the most internally incoherent response in the series to date. This is not a judgment error or a knowledge gap; it is a failure of basic reasoning consistency. The model's argument and its conclusion point in opposite directions. Combined with its concerning answers on Confined Space (#6) and Odor Pipe (#11), Gemma 1B's pattern suggests that 1B-class models may lack the reasoning capacity for domain-specific questions, even straightforward ones.

Gemma 4B's failure reveals a subtler issue: question reframing. By interpreting "odor control" as "odor reduction" rather than "odor containment," the model applied a different decision framework to the problem. The answer is coherent within that framework — circular shapes do reduce turbulence — but the framework itself is wrong for this question. This echoes the pattern observed in SCADA Cabling (#10), where models applied the right concept to the wrong implementation.

The series now contains twelve benchmarks across three difficulty tiers. Easy questions reveal small-model limitations. Mid-range questions separate models with domain knowledge from those without it. Hard questions challenge even the most capable models with multi-criteria evaluation traps. Together, these tiers provide a comprehensive view of how LLMs process engineering questions of varying complexity.

7. Methodology Notes

Non-determinism: Single run per model. Results are indicative, not absolute.

Single question: This report analyzes one question with 6 options. Random correct answer probability is 16.7%.

Parsing: DeepSeek R1's response begins with <think>. The adjusted pass rate (90.9%) accounts for this. R1's parsing behavior is tracked at 10 of 12 benchmarks (83%).

Token counts: Some Gemma models reported 0 output tokens in API metrics despite visible responses. Counts marked with ~ are estimated.

API conditions: All models accessed via Kaggle Benchmarks platform API with default temperature settings.

Series consistency: All twelve benchmarks use the same platform, the same first-character parser, and the same default temperature settings.